

Randomized-Depth Multi-Checkpoint Activation-DiFR: the verifier-compute optimization

difr-mid project notes

August 21, 2026

Abstract

Upstream Activation-DiFR verifies a provider’s forward pass by comparing Johnson–Lindenstrauss fingerprints of the hidden state feeding the unembedding, at the cost of one full-depth teacher-forced pass per audited token. We fingerprint at multiple depths and audit at a *random* depth, so expected verifier compute falls to $\mathbb{E}[d]/L$ of a full pass. Randomization is load-bearing: any fixed shallow audit depth is fully exploitable by a suffix cheater, whereas a randomized depth catches a cheat in segment i with per-token probability $\Pr_\pi[d \geq c_i]$ and escape probability decaying geometrically in the number of audited tokens. The audit distribution π is the knob trading compute against time-to-detection, and its optimum has a simple two-point structure.

1 Setup

Let M be the declared model with L transformer layers and hidden width d_{model} . For a sequence $x_{1:T}$, write $h_t^{(\ell)} \in \mathbb{R}^{d_{\text{model}}}$ for the residual-stream state at position t after layer ℓ (with $h_t^{(L)}$ taken *post* final normalization, i.e. the input to `lm_head`). By causality,

$$h_t^{(\ell)} = f_\ell(x_{1:t})$$

depends only on the prefix, so a single teacher-forced forward pass over $x_{1:T}$ yields $\{h_t^{(\ell)}\}_{t \leq T, \ell \leq L}$ in parallel. This is what makes verification cheap relative to generation: the provider decoded autoregressively in T sequential steps; the verifier replays all of them in one prefill.

Checkpoints. Fix a checkpoint set

$$\mathcal{C} = \{c_1 < c_2 < \dots < c_m = L\} \subseteq [L],$$

with $c_0 := 0$. The layers $(c_{i-1}, c_i]$ form the i -th *segment*.

Fingerprints. For each checkpoint $c \in \mathcal{C}$ the verifier publishes a seed s_c that determines a Johnson–Lindenstrauss matrix

$$P_c \in \mathbb{R}^{k \times d_{\text{model}}}, \quad (P_c)_{ij} \sim \mathcal{N}(0, \tfrac{1}{k}) \text{ i.i.d.}, \quad k \ll d_{\text{model}}.$$

The prover, during its ordinary forward pass, emits for every generated position t the fingerprint vector

$$\phi_t = (P_{c_1} h_t^{(c_1)}, \dots, P_{c_m} h_t^{(c_m)}) \in \mathbb{R}^{mk},$$

i.e. mk floats per token of overhead. Let $\tilde{h}_t^{(c)}$ denote the prover’s (possibly dishonest) state and $\tilde{\phi}_{t,c} = P_c \tilde{h}_t^{(c)}$ the shipped fingerprint at checkpoint c .

Distance and acceptance. The verifier recomputes $h_t^{(c)}$ teacher-forced and forms

$$D_{t,c} = \|P_c h_t^{(c)} - \tilde{\phi}_{t,c}\|_2.$$

By the JL lemma, $\mathbb{E} D_{t,c}^2 = \|h_t^{(c)} - \tilde{h}_t^{(c)}\|_2^2$, so $D_{t,c}$ is an unbiased low-dimensional proxy for the true activation discrepancy. Honest cross-implementation runs (different kernels, batching, dtypes) are not bit-exact, so for each checkpoint a threshold τ_c is calibrated on honest traffic at a target false-positive rate α :

$$\tau_c = \inf \left\{ \tau : \Pr_{\text{honest}} [\bar{D}_c > \tau] \leq \alpha \right\}, \quad \bar{D}_c = \frac{1}{T} \sum_{t=1}^T D_{t,c},$$

where the per-sequence mean $\bar{D}_c$ is used rather than the per-token maximum because the latter is too heavy-tailed. A checkpoint *fails* if $\bar{D}_c > \tau_c$.

2 Verifier cost

Let κ_ℓ be the cost of layer ℓ (forward, per token) and $K(\ell) = \sum_{j \leq \ell} \kappa_j$ the cumulative cost; $K(L)$ is one full pass. Upstream Activation-DiFR verifies at depth L only, costing $K(L)$ per audited token. Since $h_t^{(c)}$ does not depend on layers $> c$, a verifier that audits to depth d can stop after layer d and pay only $K(d)$, while still checking every checkpoint $c \leq d$. For uniform layers, $K(d)/K(L) = d/L$.

3 The coverage gap under a fixed audit depth

Suppose the verifier always audits to a fixed depth $d = c_j < L$. A *suffix cheater* computes layers $1, \dots, c_j$ honestly and substitutes arbitrary cheaper computation for layers $c_j+1, \dots, L$ (quantized suffix, skipped layers, another model’s head). Then $\tilde{h}_t^{(c)} = h_t^{(c)}$ for all $c \leq c_j$, so every audited fingerprint passes:

$$\Pr[\text{detect}] = \alpha \quad (\text{the false-positive rate}),$$

while the cheater saves exactly $K(L) - K(c_j)$ per token—the same compute the verifier saved. Any fixed shallow depth is therefore fully exploitable.

4 Randomized audit depth

Let the verifier draw, independently for each audited token, a depth

$$d \sim \pi, \quad \pi \in \Delta(\mathcal{C}), \quad \pi_i := \Pr_\pi[d = c_i],$$

and recompute layers $1..d$, checking all checkpoints $c \leq d$. Define the tail probability

$$\bar{\pi}_i := \Pr_\pi[d \geq c_i] = \sum_{j \geq i} \pi_j.$$

Propagation. A cheat that first deviates inside segment i (layers $(c_{i-1}, c_i]$) corrupts $h_t^{(c)}$ for every $c \geq c_i$, since each later layer is a deterministic function of the earlier state (and the corruption does not cancel except on a measure-zero set, which the calibrated threshold excludes). Hence it is caught at token t iff $d_t \geq c_i$ and the discrepancy exceeds the threshold; writing

$$p_i := \Pr[\bar{D}_{c_i} > \tau_{c_i} \mid \text{cheat in segment } i]$$

for the per-audit power (empirically $p_i \approx 1$ for quantization-level cheats),

$$\Pr[\text{caught at one audited token}] = \bar{\Pi}_i p_i.$$

Escape probability. Over N independently audited tokens,

$$\Pr[\text{escape}] = (1 - \bar{\Pi}_i p_i)^N \leq \exp(-N \bar{\Pi}_i p_i),$$

so the expected number of audited tokens to detection is $1/(\bar{\Pi}_i p_i)$, and detection is guaranteed with probability $\geq 1 - \delta$ once

$$N \geq \frac{\ln(1/\delta)}{-\ln(1 - \bar{\Pi}_i p_i)} \approx \frac{\ln(1/\delta)}{\bar{\Pi}_i p_i}.$$

Expected verifier compute.

$$\mathbb{E}_\pi[K(d)] = \sum_{i=1}^m \pi_i K(c_i) = \sum_{i=1}^m \bar{\Pi}_i (K(c_i) - K(c_{i-1})),$$

the second form (Abel summation) showing that each segment's cost is paid exactly with the probability that the audit reaches it. The relative saving over full-depth auditing is $1 - \mathbb{E}_\pi[K(d)]/K(L)$.

5 The optimization

The audit distribution π trades expected compute against time-to-detection. Two equivalent formulations:

(a) Minimize compute subject to a detection floor. Require every segment to be caught with per-token probability at least ρ :

$$\min_{\pi \in \Delta(\mathcal{C})} \sum_{i=1}^m \bar{\Pi}_i (K(c_i) - K(c_{i-1})) \quad \text{s.t.} \quad \bar{\Pi}_i p_i \geq \rho \quad \forall i.$$

Because $\bar{\Pi}_i$ is non-increasing in i and the objective weights are positive, the binding constraint is the deepest segment: the minimizer sets $\bar{\Pi}_m = \rho/p_m$ (audit to full depth with probability ρ/p_m) and places the remaining mass as shallow as the shallower constraints allow. With uniform $p_i = p$ and uniform layers, the optimum puts mass ρ/p on $c_m = L$ and $1 - \rho/p$ on c_1 , giving

$$\frac{\mathbb{E}_\pi[K(d)]}{K(L)} = \frac{c_1}{L} + \frac{\rho}{p} \left(1 - \frac{c_1}{L}\right),$$

which approaches c_1/L as $\rho \rightarrow 0$ while keeping every segment's escape probability decaying as $(1 - \rho)^N$.

(b) **Maximize worst-case detection subject to a compute budget.**

$$\max_{\pi \in \Delta(\mathcal{C})} \min_i \bar{\Pi}_i p_i \quad \text{s.t.} \quad \sum_{i=1}^m \pi_i K(c_i) \leq B.$$

This is a linear program in π ; the min is again attained at $i = m$, so it reduces to maximizing π_m under the budget, with the same two-point structure.

Cheater’s economics. A cheater saving $\sigma_i := K(L) - K(c_{i-1})$ per token by cheating from segment i onward, under a penalty Λ on detection, profits only if

$$\sigma_i \cdot N_{\text{served}} > \Lambda \cdot \left(1 - (1 - \bar{\Pi}_i p_i)^{N_{\text{audited}}}\right),$$

and since the right-hand side approaches Λ exponentially fast in N_{audited} , any $\bar{\Pi}_i p_i > 0$ makes sustained cheating unprofitable for large enough audit volume. The design choice is therefore not *whether* to audit deep, but how rarely one can afford to.

6 Summary of the identity

$$\underbrace{\sum_i \bar{\Pi}_i \Delta K_i}_{\text{verifier cost}} \quad \text{vs.} \quad \underbrace{\bar{\Pi}_i p_i}_{\text{per-token catch rate of segment } i}$$

share the same tail probabilities $\bar{\Pi}_i$: every unit of compute spent reaching checkpoint c_i buys detection probability for *every* cheat that starts at or before it, and a fixed shallow policy is exactly the degenerate point $\bar{\Pi}_m = 0$.

Notes on simplifications

- $p_i \approx 1$ matches the empirical results in this repo (fp8-KV and int4-suffix cheats flagged 100% once the audit reaches the corrupted checkpoint); keeping p_i explicit lets a measured ROC operating point be substituted.
- The two-point optimum (mass on the shallowest and deepest checkpoints) is what the LP gives under uniform p_i . Intermediate checkpoints carry mass only under heterogeneous p_i , a cheater restricted to certain segments, or a per-sequence rather than per-token depth draw.
- “Corruption propagates and does not cancel” is an assumption justified by the calibrated threshold: a cheater cannot land inside the noise floor at a later checkpoint without knowing the true state there, which requires doing the honest work up to that point.